

Causal Disentanglement for Tackling Popularity Bias in Sequential Recommendation

An-An Liu , Senior Member, IEEE, Yadong Zhao , Xin Wen , Rihao Chang , and Weizhi Nie 

Abstract—Recommender systems typically exhibit severe popularity bias, with a few highly popular items receiving excessive exposure. Most existing studies tackle this bias in static settings. However, they neglect the dynamic nature of real-world recommendation scenarios and lack a thorough analysis into the root causes of bias, which makes it challenging to accurately model and mitigate the dynamically changing popularity bias and capture genuine user preferences. To this end, we propose a causal disentanglement sequential recommendation model (CDSRec) based on time series analysis and hidden variable separation. Our model leverages Markov chains to analyze historical interaction data within sequential recommendations, capturing the dynamic variations of item popularity and user preferences. Employing causal inference, we disentangle the potential factors implicated in popularity bias. Specifically, user–item interactions are primarily driven by personalized demands and item popularity. Through empirical analysis from a temporal perspective, we reveal that popularity has both positive and negative impacts, and attribute them to stable intrinsic quality factors and dynamic external interference factors. We construct a causal directed acyclic graph to elucidate the temporal correlations among different factors. Subsequently, we utilize historical interaction sequences and item-related attributes as auxiliary information to explicitly disentangle these factors as hidden variables. By reformulating the objective function to optimize the sequential VAE framework, our model effectively mitigates the negative impact of external interference factors. Extensive experimental results on three real-world datasets demonstrate the superiority of our proposed model.

Impact Statement—In the digital age, recommender systems are essential tools for addressing information overload, with their performance directly impacting user experiences. However, popularity bias often results in homogeneous recommendations, and existing methods struggle to address this bias in real-world dynamic scenarios. Our proposed CDSRec model innovatively leverages a causal inference framework and Markov chains to analyze time series information, disentangle latent factors that affect popularity bias, and effectively mitigate its negative effects. Experiments demonstrate that CDSRec surpasses state-of-the-art

methods in capturing genuine user preferences and improving recommendation accuracy. This approach is broadly applicable to platforms like e-commerce and online streaming media, enhancing user satisfaction and advancing the development of recommendation technologies.

Index Terms—Causal disentanglement, popularity bias, sequential recommendation, time series analysis.

I. INTRODUCTION

IN the digital age, recommender systems have become an essential tool for addressing information overload problem [1], widely used across platforms such as e-commerce [2] and online streaming services [3]. However, these systems often suffer from popularity bias [4], where interaction data usually exhibits an unbalanced or long-tail distribution, with a small number of highly popular items receiving excessive exposure to users, whereas most items are seldom exposed. Models trained on these unbalanced data tend to amplify these biases, creating a feedback loop that leads to problems such as Matthew effect [5] and echo chambers [6]. As a result, the recommendation results become increasingly homogeneous [7], impairing the system’s ability to accurately understand users’ actual preferences and subsequently degrading the user experience.

Most existing studies tackle popularity bias in static environment settings, assuming that popularity bias and user preferences remain constant over time [8], and evaluate models through single-round recommendation [9]. However, this setup diverges from real interaction scenarios. In the real world, the recommendation process is inherently dynamic, with both item popularity and user preferences evolving over time. This dynamic nature can be reflected in the temporal information such as user–item interaction sequences and rating feedback. When addressing the problem of popularity bias, lacking the processing and analysis of temporal information prevents models from promptly perceiving the shifts in popularity, thereby failing to effectively mitigate the popularity bias in dynamic scenarios. This limitation leads to models that cannot accurately capture true user preferences, ultimately resulting in reduced accuracy in recommendation results.

Unlike traditional recommender systems, sequential recommendation focuses on analyzing users’ historical behavior sequences to predict their future interests [10]. Recently, sequential debiasing methods have attracted attention from researchers. The latest works primarily employ contrastive learning and adaptive augmentation to filter biased samples

Received 29 December 2024; revised 11 April 2025; accepted 28 May 2025. Date of publication 4 June 2025; date of current version 31 December 2025. This work was supported in part by the National Natural Science Foundation of China under Grant U21B2024 and Grant 62272337. This article was recommended for publication by Associate Editor Sriparna Saha upon evaluation of the reviewers’ comments. (Corresponding authors: Xin Wen; Weizhi Nie.)

An-An Liu, Yadong Zhao, Xin Wen, and Weizhi Nie are with the School of Electrical and Information Engineering, Tianjin University, Tianjin 300072, China (e-mail: anan0422@gmail.com; 2023234200@tju.edu.cn; wenxin113@tju.edu.cn; weizhinie@tju.edu.cn).

Rihao Chang is with the School of Microelectronics, Tianjin University, Tianjin 300072, China (e-mail: changrihao@tju.edu.cn).

Digital Object Identifier 10.1109/TAI.2025.3575554

[11], [12]. However, these methods still overlook the dynamic nature of real-world scenarios, lacking an in-depth analysis of the root causes of popularity bias and a timely capture of its dynamic changes. To address this issue, we attempt to utilize time-series data in sequential recommendation to model and mitigate the dynamically changing popularity bias. This allows the model to better distinguish true user preferences from popularity-driven interactions. To this end, we analyze the causes of popularity bias from a temporal perspective. In real-world scenarios, user-item interactions are shaped by two primary factors: user interest and item popularity. Considering temporal information, we recognize that popularity has both positive and negative impacts. Items with enduring popularity may possess higher intrinsic quality, whereas items that gain short-term popularity due to external factors often fail to meet user expectations.

To validate our viewpoint, we conduct empirical analysis on relevant datasets to reveal time-sensitive correlations between item popularity and user satisfaction. Item popularity is primarily reflected in the frequency of interactions, such as clicks, views, and purchases. User satisfaction can be indicated by the ratings assigned to items, which indirectly reflect user preferences. This analysis leads to the following two observations: 1) in the long term, the popularity of items generally shows a positive correlation with their average ratings. This implies that popularity bias is not entirely harmful. Items with long-term popularity often possess higher intrinsic quality, which leads to sustained user preference; and 2) over a short period, inconsistencies are observed between item popularity and user satisfaction. This may be due to some external factors that lead to items receiving widespread attention in the short term, increasing the conformity influence on user behavior and causing a divergence from personal true preferences. This reflects the negative impact of popularity bias. The empirical analysis results align with real-world patterns. Some items attain enduring popularity due to their inherent excellent characteristics, such as good quality; whereas others spread widely in the short term because of some external factors, like the star effect. Related to temporal information, these factors are crucial in tackling the negative impacts of popularity bias.

Based on the above analysis of popularity bias, we categorize the latent factors that affect recommendation results into three parts: 1) personalized demand factors: this hidden variable reflects the match degree between items and user preferences, which can be inferred from the user's historical interaction sequence; 2) inherent quality factors: this variable represents the beneficial aspects of popularity, primarily determined by factors such as item intrinsic quality that are not easily susceptible to external influences. It can be reflected from the item's long-term interaction and rating feedback; and 3) external interference factors: this variable represents harmful popularity factors that driven by external influences such as trending social events. It captures item features that experience transient popularity but lack long-term value. Popularity and rating information dynamically change and show a negative correlation, adversely affecting the recommendation accuracy and user experience. By designing models to disentangle these factors and leverage

the first two for recommendations, we can mitigate the adverse effects of external interference and capture true user preferences in dynamic scenarios.

In this article, we propose the causal disentanglement sequential recommendation model (CDSRec), a novel approach designed to tackle the challenges of popularity bias in dynamic scenarios. We emphasize the importance of temporal information in addressing the popularity bias problem. By conducting time-series analysis on historical interaction data, we capture the dynamic shifts of item popularity and user preferences. Through empirical analysis from a temporal perspective, we identify the factors influencing popularity bias. Then we construct a causal directed acyclic graph to elucidate the temporal correlations among different factors. Based on causal inference and hidden Markov model, we utilize interaction sequences and item-related attributes as auxiliary information to explicitly disentangle these factors as hidden variables. By reformulating the objective function to optimize the sequential variational autoencoder (VAE) framework, our model effectively mitigates the negative impact of external interference factors. To demonstrate the effectiveness of our proposed model, we conduct thorough experiments on the KuaiRec, Douban-Movie, and Amazon-CDs datasets. The results indicate that our approach achieves significant improvements over current state-of-the-art methods.

The primary contributions of this article can be summarized as follows:

- 1) We innovatively address popularity bias in sequential recommendation by explicitly modeling its temporal dynamics, aligning better with the real-world scenarios.
- 2) We propose the CDSRec model, which employs causal inference to disentangle the potential factors affecting popularity bias. By optimizing sequential VAE framework, we effectively mitigate the negative impact of external interference factors.
- 3) Experimental results on three real-world datasets demonstrate the improvement in recommendation performance of our proposed method.

The rest of this article is structured as follows. Section II reviews the related work. Section III conducts an empirical analysis and briefly describes the model background. Section IV details our proposed model and learning method. The experimental results and analysis are presented in Section V. Lastly, Section VI concludes our work and discusses future research directions.

II. RELATED WORK

In this section, we will divide the content into two parts to introduce the sequential recommendation system and the current mainstream methods for addressing popularity bias, respectively.

A. Sequential Recommendation

Compared to static recommendation systems, sequential recommendation typically generates user representations based on interaction sequences [13]. Markov chain-based methods [14], [15] predict the next item by modeling k -order

interactions, laying the foundation for subsequent actions. With the advancement of deep learning, recurrent neural networks (RNNs) and their variants, such as LSTM and GRU, have been extensively applied to sequential recommendation tasks [16]. For instance, Hidasi et al. [17] designed specialized ranking loss functions and optimization techniques for RNN and GRU modules. The introduction of hierarchical structures [16] and attention mechanisms [18], [19] has similarly enhanced recommendation performance. Kang et al. [20] utilized transformer layers to dynamically assign weights, adapting to variations in users' historical interactions. Beyond RNNs, convolutional neural networks (CNNs) have also been employed in sequential recommendation tasks [21], [22]. The AdaMCT model proposed by Jiang et al. [23] combined CNN and transformer to simultaneously capture both long-term and short-term user preferences.

In recent years, graph neural networks (GNNs) have made significant strides in processing data with graph structures, leading to numerous studies applying it to sequential recommendation tasks [24]. For example, SR-GNN was the first to employ gated GNNs to capture complex item relationships in session-based scenarios [25]. DGSr utilizes dynamic GNNs to capture the evolving collaborative signals between user sequences [26]. Similarly, SelfGNN [27] integrates self-supervised learning with graph neural networks to capture patterns of user behavior over different time spans.

Overall, research in sequential recommendation encompasses a variety of frameworks, from Markov chains to deep learning models, each optimized for different characteristics of user behavior.

B. Popularity Bias in Recommendation

Popularity bias refers to the inclination of recommendation systems to favor popular items or users, potentially leading to suboptimal recommendations and fairness problems. It has attracted widespread attention from researchers in recent years. Current research methods can be roughly divided into the following categories.

Methods based on inverse propensity scoring (IPS) are widely utilized to adjust the recommendation outcomes by estimating the propensity weights of items [28], [29]. These methods mitigate popularity bias by assigning higher weights to less popular items and reducing the weights of more popular ones. Similarly, Steck et al. [30] proposed directly adjusting the predicted scores to increase the exposure of less popular items, thereby achieving a more balanced set of recommendations. However, the variance of the IPS method is typically large [31], leading some researchers to employ training methods based on regularization and ranking adjustment. For instance, model regularization on the training [32] or add regularization terms to the loss function [33]. Chen et al. [34] proposed the ESAM model which introduces attribute correlation alignment and two regularization strategies to reduce the distribution discrepancy between displayed and nondisplayed items. Abdollahpouri et al. [35] fine-tuned the ranking of recommendations for users by considering their tolerance for well-liked items.

Additionally, with the development of causal inference technology, it has been widely applied in recommendation systems [36]. Some methods have been proposed to reduce popularity bias. PDA [37] mitigates popularity bias in model training and enhances recommendation accuracy by adjusting popularity in the inference phase. DICE [38] uses causal inference and multitask learning to tackle data bias and improve recommendation interpretability by separating user interest embeddings from conformity behavior. Wang et al. [39] proposed CaDSI, a causal disentanglement approach that uncovers unbiased, semantic-aware user intents to improve recommendation system accuracy. Liu et al. [40] introduced an adaptive recommendation model FAiR, which employs causal disentanglement to address popularity bias, aiming to ensure fairness in recommendations for both users and items.

C. Sequential Debaised Methods

Earlier studies on popularity bias primarily focus on static recommendation tasks, which only represent a simplified subprocess of the dynamic real-world process [29]. Recently, some work has attempted to leverage sequential user-item interaction information to better simulate real-world scenarios and address the issue of popularity bias. For example, DCRec [11] integrates contrastive learning with adaptive augmentation to distinguish between user conformity and genuine interest, thereby mitigating popularity bias in sequential patterns. The DRO-based method [41] constructs an exposure simulator and applies distributionally robust optimization to adjust system exposure, preventing the overestimation of user preferences. Additionally, SimDCL [12] enhances contrastive learning by improving representation consistency and filtering biased samples, while STOSA [42] employs stochastic embeddings and Wasserstein self-attention to improve recommendation robustness.

Nevertheless, existing studies still face limitations in addressing popularity bias. First, the real-world recommendation environments are inherently dynamic [13]. Specifically, user preferences may evolve due to changes in life circumstances or aging, while item popularity can fluctuate suddenly due to promotions or trending social events, with some items gradually losing their appeal over time. This necessitates recommendation systems that can promptly capture these changes. If the system blindly recommends popular items that do not align with user preferences, it will not only exacerbates the effects of popularity bias but also severely reduces user satisfaction. Existing methods perform poorly in capturing the dynamic changes in user preferences and item popularity. Second, current sequential debiasing approaches fail to thoroughly analyze the underlying causes of item popularity, making it difficult to effectively mitigate the negative impact of popularity on the recommendation process. Our approach provides a new perspective for alleviating popularity bias in sequential recommendation.

III. PRELIMINARIES

A. Empirical Study on Popularity Bias

In this section, we conduct an empirical analysis of popularity bias in sequential recommendation. We select the

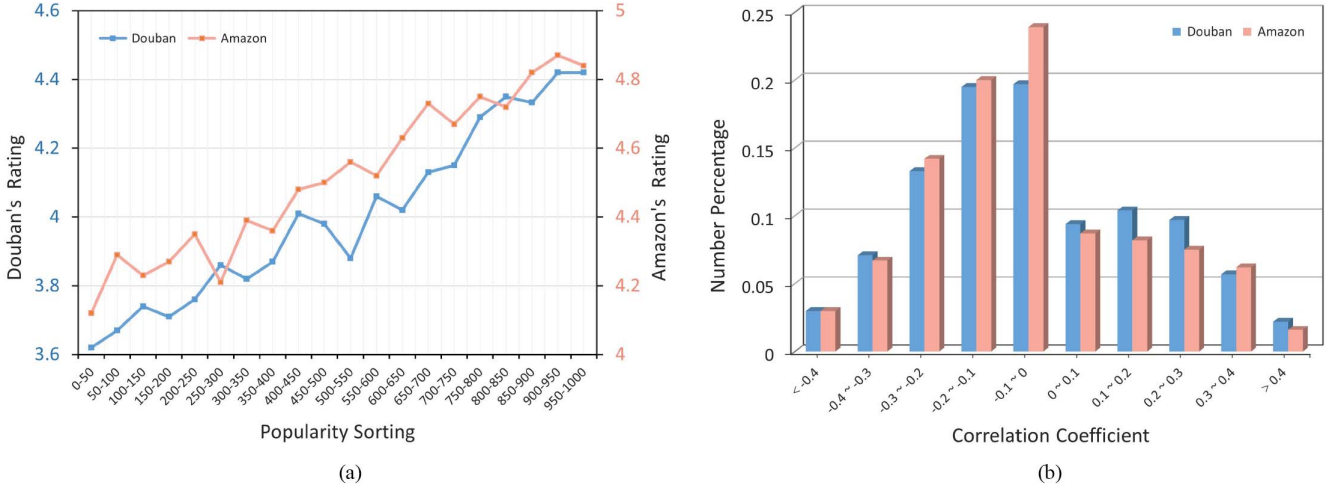


Fig. 1. Subplot (a) displays the trends in item popularity and their corresponding average ratings among different groups for the Douban-movie and Amazon-CDs datasets. Subplot (b) depicts the distribution of Pearson correlation coefficient between temporary popularity changes and average ratings over a 90-day period. (a) Average rating trends with popularity. (b) Coefficient distribution on two datasets.

Douban-Movie and Amazon-CDs datasets, which contain comprehensive user-item interaction timestamps and item ratings. Higher ratings typically indicate greater user satisfaction with the items. Building on existing work [37], we define the popularity of item i as the product of user coverage c_i^T and interaction frequency n_i^T . Here, c_i^T denotes the proportion of users interacting with item i within the time frame $t \leq T$, and n_i^T represents the average interaction frequency over the same period. We randomly select 1000 items from each dataset and calculate their popularity, ranking them accordingly. The items were categorized into 20 levels based on their popularity, with each level containing an equal number of items. Lower levels indicate lower popularity. Additionally, we calculate the average rating for each group of items, with results illustrated in Fig. 1(a).

Furthermore, we explore the relationship between the temporary popularity and item ratings. Inspired by Zhang et al. [37], we define the temporary popularity P_i^t of item i at time t as follows:

$$P_i^t = c_i^t \cdot n_i^t - c_i^{t_o} \cdot n_i^{t_o}, \quad t_o < t \quad (1)$$

where we set a 90-day baseline (i.e., $t - t_o = 90$) and determine the average rating over the corresponding time interval. Then calculate the Pearson correlation coefficient between them. Fig. 1(b) illustrates the distribution of coefficients across the two typical datasets. Through experimental analysis, we have arrived at the following two observations:

Observation 1: In the long term, the popularity of items generally shows a positive correlation with their average ratings.

Fig. 1(a) indicates that as item popularity increases, the average rating shows a general upward trend, suggesting a positive correlation between them. Higher ratings reflect greater user acceptance, which implies that popularity is not entirely detrimental. Items with enduring popularity might be favored due to their higher intrinsic quality.

Observation 2: Over a short period, there may be inconsistencies between item popularity and user satisfaction.

It is noteworthy that although the overall ratings in Fig. 1(a) are upward, the curve exhibits significant fluctuations, indicating that certain items may be popular but not well-received by users. Additionally, the Pearson correlation coefficients shown in Fig. 1(b) reveal that a considerable number of items demonstrate a weak or even negative correlation between their temporary popularity and ratings. In the short term, items may gain widespread attention due to external factors, leading to increased conformity influence on user behavior, which diverges from genuine user preferences. This highlights the harmful effects of popularity bias.

Based on the analysis above, we make the following assumptions: when addressing the problem of popularity bias in sequential recommendation, it is essential to consider not only the personalized needs of users, but also the item inherent quality and external interference factors. The former is a stable and static attribute, but the latter shows dynamic feature that may lead to short-term fluctuations in user behavior. Our goal is to use time-series information to disentangle these factors, extracting beneficial information related to item quality while mitigating the adverse impacts of external interference.

B. Concept of Structural Causal Model

We introduce the concept of structural causal model (SCM). SCM is a framework for describing causal relationships between variables, typically represented by a directed acyclic graph (DAG) [43]. In this representation, nodes depict variables, and directed edges depict causal relationships. Each node has a structural equation that shows how its value is influenced by its parent nodes. Specifically, the DAG is defined as $G := (N, E)$ with N and E denote the node set and edge set, respectively. The arrow $a \rightarrow b$ ($a, b \in N$) indicates that a has a direct causal effect on b , influencing its probability distribution. For the node set $N := \{n_1, \dots, n_k\}$, the generative process is defined

as $n_i \leftarrow f_i(P(n_i))$, where f_i denotes the causal mechanism, describing how the parent nodes $P(n_i)$ influence the value of n_i .

The SCM allows for the identification of confounding bias, where correlations between variables may arise from a third variable rather than direct causation. For instance, in the causal graph $n_1 \leftarrow n_2 \rightarrow n_3$, the correlation between n_1 and n_3 is due to their shared confounder n_2 , rather than a direct causal relationship from n_1 to n_3 .

IV. METHODOLOGY

A. Problem Definition

Suppose that we have a sequential recommendation system with $\mathcal{U} = \{u_1, u_2, \dots, u_{|\mathcal{U}|}\}$ and $\mathcal{I} = \{i_1, i_2, \dots, i_{|\mathcal{I}|}\}$ representing the sets of users and items, respectively. Denote $\mathcal{D}_t = \{(\theta_{u,i}, t)\}$ as the historical interaction data, where $\theta_{u,i}$ is a multihot matrix. Data is collected chronologically and organized into a list of tuples. For user u_n and item i_m , $\theta_{u_n, i_m} = 1$ indicates that there was an interaction, such as browsing, before time t . While $\theta_{u_n, i_m} = 0$ signifies no interaction during that period. To clarify the model construction, we use $U_t = \{(\theta_{u_n, i_k}, t)\}_{1 \leq k \leq |\mathcal{I}|}$ to represent the interaction data of user u_n with various items before time t . Similarly, $I_t = \{(\theta_{u_k, i_m}, t)\}_{1 \leq k \leq |\mathcal{U}|}$ denotes the interaction of item i_m with users. Auxiliary variable B_t captures the attribute information, such as item ratings and temporary popularity. S_t reflects the predicted recommendation score for items. We disentangle the dynamic influence related to popularity bias into three hidden variables x_t , y_t , and z_t , which represent personalized demand factors, inherent quality factors, and external interference factors, respectively.

The problem lies in how to utilize historical interaction data and attribute information to mitigate the adverse effects of popularity bias and make accurate recommendations, i.e., $p(S_t | o_{t_1:t_2})$ with $o \in \{U, I, B\}$ and $t_1 < t_2 < t$. In this section, we will provide a detailed overview of the causal disentanglement sequential recommendation model and elucidate the method of disentangling hidden variables within this model.

B. Construction of the Causal Relationship

To better understand the debiasing structure of our model, we use causal graphs for qualitative analysis. We first introduce the traditional recommendation model, as shown in Fig. 2(a), where U and I represent user and item nodes, respectively, and S denotes the model prediction result. Typical backbone models, such as BPR-MF [44] and LightGCN [45], do not consider the impact of popularity bias.

On this basis, we incorporate the debiasing process for popularity bias. General causal debiasing models can be simply summarized as shown in Fig. 2(b). Node x represents the match degree between users and items, and node p denotes the item popularity score. To eliminate the adverse impact of popularity bias on model training, an intervention is applied to the recommended item to remove the influence of popularity p .

To address the limitations of existing methods in analyzing the causes of popularity bias and capturing its dynamic changes,

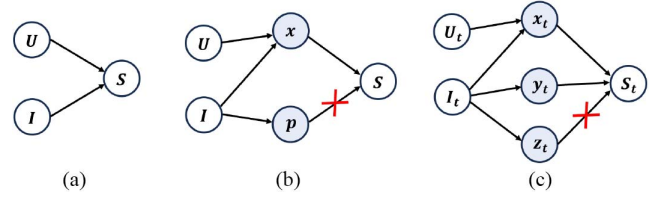


Fig. 2. Causal graphs of: (a) traditional models; (b) general debiasing models; and (c) our proposed CDSRec model. U, user; I, item; S, prediction result; p, popularity; x, personalized demand factors; y, inherent quality factors; and z, external interference factors.

we propose an improved solution, as shown in Fig. 2(c). Based on the empirical analysis conclusions from previous sections, we conduct a disentanglement analysis of the factors affecting item popularity and introduce a temporal modeling process. Specifically, we use node y_t to represent intrinsic quality factors, which signify the positive impact of popularity. This node is derived from the average rating feedback of items, as item ratings generally reflect user satisfaction and item quality. The node z_t represents external interference factors that account for the negative impact of popularity bias. These factors are characterized by rapid fluctuations in item popularity and exhibit a negative correlation with rating changes. Therefore, we calculate the item's temporary popularity using (1) and compute its Pearson correlation coefficient with rating changes as auxiliary information.

For the personalized demand factor x_t , we follow general debiasing models and utilize the typical backbone model BPR-MF to obtain basic matching score between users and items. Furthermore, considering the dynamic nature of item popularity and user preferences in real-world scenarios, we fully leverage the temporal information of sequential recommendation. The disentangled hidden variables are propagated across different time steps, enhancing the recommendation accuracy and improving the alignment with real-world requirements.

C. Structure of the CDSRec Model

To facilitate a better understanding of the implementation details of our proposed model, we make supplementary modifications to the causal graph. As shown in Fig. 3, we use the auxiliary variable B_t to represent the attribute information that guides the disentanglement of hidden variables. This includes data such as item rating feedback and temporary popularity. Given that the VAE framework focuses on the decoding and reconstruction of observed data, we replace the original input variables with the reconstructed observed variables U_t and I_t , which represent the historical interaction sequences of users and items, respectively. Inspired by the structural causal model [43] and causal hidden Markov model [46], We define the structural equations of the CDSRec model shown in the causal DAG, with the part for hidden variables $h \in \{x, y, z\}$ given by

$$x_t \leftarrow f_x(B_t, x_{t-1}), \quad y_t \leftarrow f_y(B_t, y_{t-1}), \quad z_t \leftarrow f_z(B_t, z_{t-1})$$

where $f_n := (p_a(n))$ denotes the causal relationships for node n , $p_a(n)$ represents the parent nodes of n . These hidden variables are influenced by the auxiliary variable B_t and depend

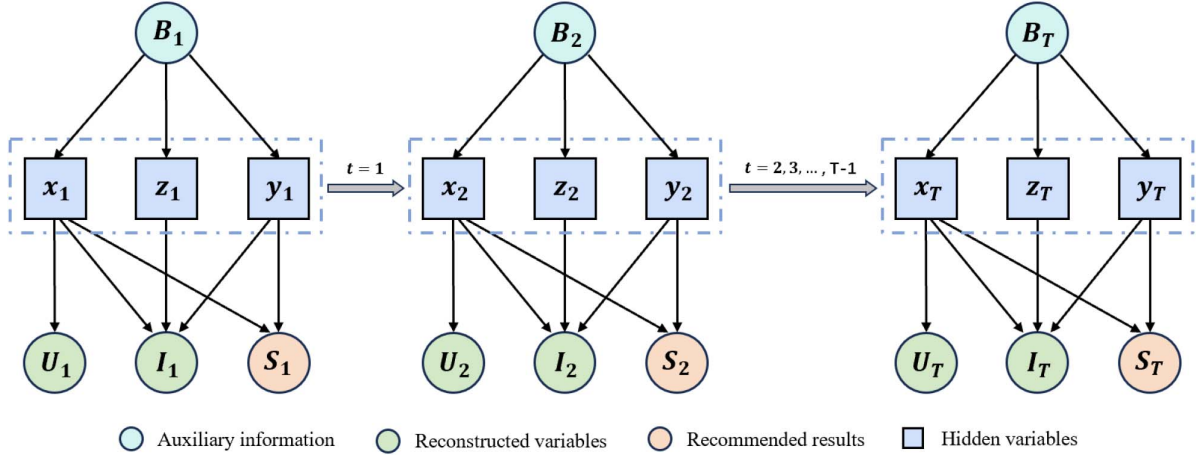


Fig. 3. Causal DAG for sequential recommendation. Hidden variables x_t (personalized demand factors), y_t (inherent quality factors), and z_t (external interference factors) are influenced by the auxiliary variable B_t . At each time step t , these variables are used to reconstruct the item interaction I_t . The objective recommendation score S_t is determined by both x_t and y_t , and the user interaction U_t is uniquely connected by x_t .

on data from the previous time step. They are propagated and updated at each time step to reflect changes in characteristics while also contributing to the reconstruction of observed variables. The structural equations for the reconstructed observed variables $o \in \{U, I, S\}$ are defined as

$$U_t \leftarrow f_U(x_t), \quad S_t \leftarrow f_S(x_t, y_t), \quad I_t \leftarrow f_I(x_t, y_t, z_t).$$

Consistent with the previous analysis of causal relationships, reconstructing observed variables is the reverse process of input generation. Specifically, the hidden variables x_t , y_t and z_t are all related to item interaction I_t . And the target recommendation score S_t is influenced by x_t and y_t , whereas user interaction U_t is directly related only to x_t . Our model aims to disentangle the key hidden variables that benefit the recommended results (x_t, y_t) from the external interference factors z_t . This separation helps us better mitigate the negative impact of popularity bias, thereby improving the recommendation accuracy.

Based on the hidden Markov condition [47], the causal relationship between the observed variables and hidden variables can be factorized through the joint distribution as follows:

$$p(h_{\leq T}, o_{\leq T}, S_{\leq T}) = \prod_{t=1}^T \left[p_{\psi}(h_t | h_{t-1}, B_t) \cdot p(U_t | x_t) \cdot p(S_t | x_t, y_t) \cdot p(I_t | h_t) \right] \quad (2)$$

where $p(U_t | x_t)$, $p(S_t | x_t, y_t)$ and $p(I_t | h_t)$ are learned within the generative network. $p_{\psi}(h_t | h_{t-1}, B_t)$ represents the prior distribution. We hope to explore the dynamic evolution of hidden variables from the prior distribution. However, due to the complexity of direct computation, we adopt the posterior distribution $q_{\phi}(h_{\leq T} | S_{\leq T}, o_{\leq T})$ for approximation. This approach can infer the distribution of hidden variables more efficiently. To better capture these causal relationships, we propose a framework based on the sequential VAE, including prior and posterior network, as shown in Fig. 4. The figure also vividly illustrates the generation and updating process of observed variables, extracting sequential information from both interaction

and attribute data. The specific model network framework is introduced as follows.

1) *Prior*: The prior network takes the observed variable B_t at the current time step and the hidden variables h_{t-1} from previous time step as inputs. For $h \in \{x, y, z\}$ at time step t , the prior distribution $p_{\psi}(h_t | h_{t-1}, B_t)$ is distributed as $\mathcal{N}(\mu(h_{t-1}, B_t), \Sigma(h_{t-1}, B_t))$. They are parameterized by a gated recurrent unit (GRU) [48] to capture long-term dependencies in sequential data. Specifically, the GRU utilizes the reset gate and update gate to selectively update information. The propagation process is as follows:

$$\begin{aligned} r_t &= \sigma(W_r \cdot [B_t, h_{t-1}] + b_r) \\ h'_t &= \tanh(W_h \cdot [B_t, r_t * h_{t-1}] + b_h) \\ v_t &= \sigma(W_v \cdot [B_t, h_{t-1}] + b_v) \\ h_t^p &= (1 - v_t) * h_{t-1} + v_t * h'_t \end{aligned} \quad (3)$$

where r_t and v_t represent the states of reset gate and update gate, respectively. σ denotes the sigmoid activation function, and $*$ signifies element-wise multiplication. Prior hidden state h_t^p is determined by the update gate state v_t and the candidate hidden state h'_t . The output of the GRU is processed through two independent fully connected (FC) layers to derive the mean and log variance vectors of the hidden variables.

2) *Posterior*: Within the Bayesian framework, the posterior distribution generally captures the conditional probability of hidden variables based on the observed data. Considering that we expect q_{ϕ} to align with the prior assumption, we make it reparameterized in the same manner as p_{ψ} , that is, $q_{\phi}(h_t | o_t, h_{t-1}) \sim \mathcal{N}(\mu(o_t, h_{t-1}), \Sigma(o_t, h_{t-1}))$. Drawing on the analysis of the prior distribution p_{ψ} , we present the posterior formulation as follows:

$$q_{\phi}(h_{\leq T} | S_{\leq T}, o_{\leq T}) = \frac{\prod_{t=1}^T [q_{\phi}(h_t | o_t, h_{t-1}) \cdot q_{\phi}(S_t | x_t, y_t)]}{q_{\phi}(S_{\leq T} | o_{\leq T})} \quad (4)$$

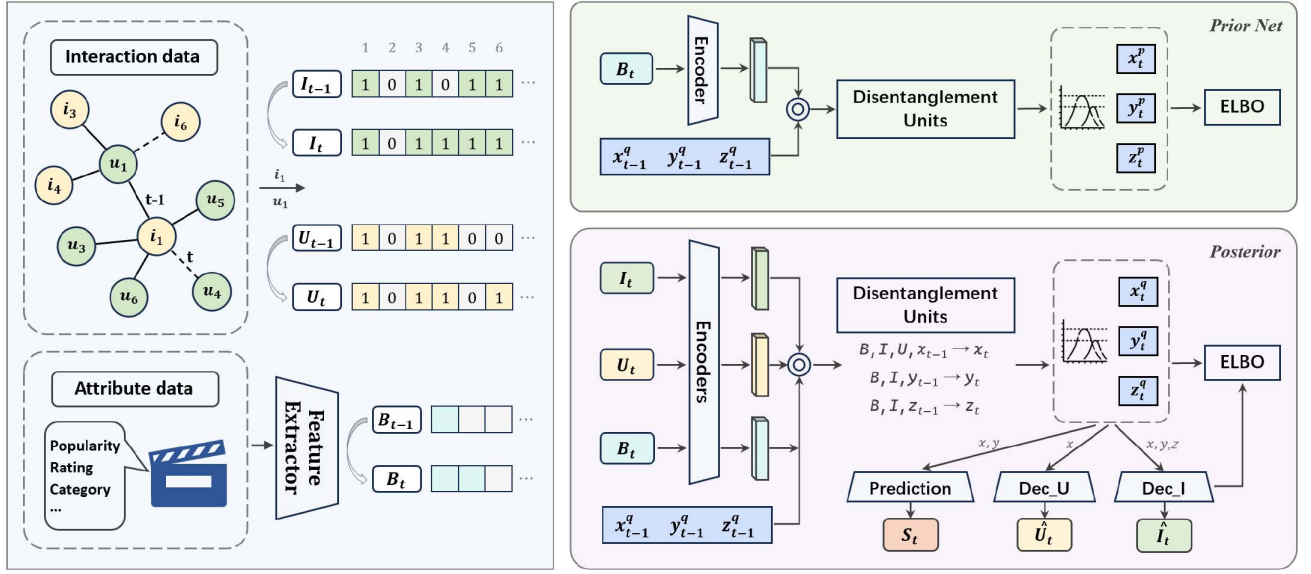


Fig. 4. Model network framework of CDSRec. The left figure illustrates the generation process of model variables, extracting sequential information from interaction and attribute data. The right figure demonstrates the disentanglement process of the prior and posterior networks. The prior network processes the concatenated vectors, which include features encoded from auxiliary data B_t and the previous hidden state $(x_{t-1}^q, y_{t-1}^q, z_{t-1}^q)$, through disentanglement units. These units comprise the GRU followed by two independent FC layers, which output the mean and log variance vectors for generating the prior hidden variables. The posterior network similarly inputs the concatenated vectors into FC layers to derive the mean and log variance vectors for posterior hidden variables. Then they are decoded to reconstruct observed variables and predict recommended items.

Directly solving the posterior distribution of latent variables is challenging. To facilitate analysis and computation, we approximate the posterior distribution with a known simple distribution, as formulated below

$$h_t^q = W_h^q \cdot [o_t, h_{t-1}] + b_h^q. \quad (5)$$

Here, h_t^q represents the posterior hidden state at time t . Based on the previous analysis of causal relationships among variables, the input o_t to the posterior network varies accordingly for different hidden variables $h \in \{x, y, z\}$. When h is x , the input o_t consists of the concatenated vectors $[B_t, I_t, U_t]$, but for y and z , the input is $[B_t, I_t]$. For the generated hidden variables x_t^q, y_t^q and z_t^q , we follow a similar approach to the prior network, employing Fully Connected layers to derive their mean and log variance vectors, which are subsequently used to reconstruct the observed variables.

3) *Generations*: In our model, the prior and posterior encoders employ a GRU structure and linear transformation to obtain the distributions of the hidden variables. These distributions are used in the generative network to reconstruct the observed variables. Specifically, we employ the variables x_t^q, y_t^q and z_t^q to reconstruct the item interaction data $\hat{I}_t$, and we utilize x_t^q to reconstruct the user behavior data $\hat{U}_t$. Additionally, the variables x_t^q and y_t^q are used to predict the recommendation score S_t . During the training process, we leverage the generated reconstructed variables $\hat{I}_t$ and $\hat{U}_t$ to approximate the true observed data I_t and U_t , ensuring that the generated variables closely resemble the actual data. This allows the model to better learn the underlying data characteristics and make accurate recommendations.

D. Learning Method

To optimize the model, we introduce the evidence lower bound (ELBO) as the objective function. The optimization goal is to maximize the ELBO, which consists of the reconstruction loss and KL divergence. Specifically, the reconstruction loss measures the similarity between the model-generated data and the real data. By maximizing the expected log-likelihood of the reconstructed data, the model is optimized to make the generated data as close as possible to the actual observed data. The expression for the reconstruction loss is presented below

$$L_1 = E_{q_\phi(h_{\leq T}|S_{\leq T}, o_{\leq T})} \log[p_\psi(S_{\leq T}, o_{\leq T})]. \quad (6)$$

The KL divergence measures the discrepancy of the hidden variables between the posterior distribution q_ϕ learned by the model and the prior distribution p_ψ . Minimizing the KL divergence ensures that the distribution of hidden variables aligns more closely with the prior assumptions. The formula for KL divergence is as follows:

$$L_2 = -D_{kl}(q_\phi(h_{\leq T}|S_{\leq T}, o_{\leq T}) \parallel p_\psi(h_{\leq T}|S_{\leq T}, o_{\leq T})). \quad (7)$$

The ELBO can be reformulated as $\mathbb{E}_{p(S_{\leq T}, o_{\leq T})}(L_1 + L_2)$. By expanding the terms based on the definition of the KL divergence, we derive the simplified ELBO as

$$\mathbb{E}_{p(S_{\leq T}, o_{\leq T})} \left[E_{q_\phi(h_{\leq T}|S_{\leq T}, o_{\leq T})} \log \left(\frac{p_\psi(h_{\leq T}, S_{\leq T}, o_{\leq T})}{q_\phi(h_{\leq T}|S_{\leq T}, o_{\leq T})} \right) \right]. \quad (8)$$

Substituting the prior (2) and the posterior (3), the ELBO can be reformulated as

$$\mathbb{E}_{p(S_{\leq T}, o_{\leq T})} \left[\sum_{t=1}^T \mathcal{L}_{q_\phi, p_\psi}^t + \log q_\phi(S_{\leq T}|o_{\leq T}) \right] \quad (9)$$

TABLE I
STATISTICS OF THE EXPERIMENTAL DATASETS

Datasets	# Users	# Items	# Interactions	Date
KuaiRec	7176	10 728	12 529 587	2020.7-2020.9
Douban-Movie	47 165	25 631	7 116 014	2010.1-2017.3
Amazon-CDs	19 210	38 941	573 126	2008.1-2014.7

where $\mathcal{L}_{q_\phi, p_\psi}^t$ is given by

$$\mathcal{L}_{q_\phi, p_\psi}^t = \mathbb{E}_{q_\phi}(h_t | o_t, h_{t-1})[(l_1 + l_2 + l_3)]. \quad (10)$$

In (9), l_1 represents the reconstruction loss, with l_2 , l_3 respectively denoting the KL divergences for the predictive distribution and the hidden variables. The specific forms of l_1 , l_2 and l_3 are given below

$$\begin{aligned} l_1 &= \log(p_\psi(U_t | x_t) \cdot p_\psi(I_t | h_t)) \\ l_2 &= \log \frac{p_\psi(S_t | x_t, y_t)}{q_\phi(S_t | x_t, y_t)} \\ l_3 &= \log \frac{p_\psi(h_t | h_{t-1}, B_t)}{q_\phi(h_t | h_{t-1}, o_t)}. \end{aligned}$$

We parameterize $p_\psi(S_t | x_t, y_t)$ as $q_\phi(S_t | x_t, y_t)$, simplifying l_2 to zero. Additionally, in (8), based on the probabilistic derivations, we have $q_\phi(S_{\leq T} | o_{\leq T})$ expressed as

$$\int \prod_{t=1}^T [q_\phi(h_t | h_{t-1}, o_t) \cdot q_\phi(S_t | x_t, y_t)] dh_0 \dots dh_T. \quad (11)$$

V. EXPERIMENTS

In this section, we conduct experiments on real datasets and compare with baseline models to evaluate the effectiveness of our proposed CDSRec. Our experiments aim to address the following three key questions:

- 1) *RQ1*: Does CDSRec perform better than SOTA methods in sequential recommendation tasks?
- 2) *RQ2*: Can the model effectively disentangle the factors influencing popularity bias and eliminate their negative impacts?
- 3) *RQ3*: Is it beneficial to incorporate temporal information when addressing the problem of popularity bias?

A. Experimental Setup

In this section, we will provide a detailed description of our experimental setup, including datasets, baseline models, implementation details, and evaluation metrics.

1) *Datasets*: We select three public datasets for our experiments: KuaiRec [49], Douban-Movie [50], and Amazon-CDs [51]. These datasets are considered representative, which are collected from authentic application scenarios. Statistical details about them are presented in Table I.

- a) *KuaiRec*: This is a real-world dataset with millions of interaction records from the short video platform Kuaishou, including video types, interaction information, and ratios of viewing time to total duration. Drawing on [8], we

adopt the watch ratio to quantify item ratings, as the viewing duration reflects the degree of user preference. For ratios exceeding 5, we cap it at 5 to maintain a rating scale between 0 and 5.

- b) *Douban-movie*: This is collected from the popular review website Douban. It includes movie-related attribute information, item ratings and rating timestamps, with millions of data spanning a long period. Ratings are scaled from 1 to 5, and we use the rating timestamps to represent interaction times.
 - c) *Amazon-CDs*: The dataset, gathered from the Amazon platform, encompasses a vast array of information including user ratings on a scale from 1 to 5, transaction information, and timestamps, etc. Its extensive temporal coverage makes it an ideal choice for analyzing long-term user behavior and dynamic preferences.
- 2) *Baselines*: We compare the proposed CDSRec with three sets of baseline methods. The first group consists of mainstream static recommendation system methods, including BPR-MF and LightGCN. The second group focuses on recommendation methods that address popularity bias using causal inference, such as CausE, DICE, and PDA. The third group comprises advanced sequential recommendation methods, including BERT4Rec, DT4SR, and STOSA.
- a) *BPR-MF* [44] is a recommendation algorithm that employs matrix factorization to learn latent features of users and items, utilizing Bayesian methods to rank items based on users' historical behavior.
 - b) *LightGCN* [45] is a graph convolutional network model for recommendation systems. It optimizes the embedding of users and items through linear information propagation within their interaction network, and uses the weighted sum of these embeddings as the final output.
 - c) *CausE* [52] learns representations of users and items by combining biased recommendation data with unbiased random exposure data, aiming to predict user behavior under a completely random recommendation strategy.
 - d) *DICE* [38] employs causal inference and multitask learning to address data bias problems. It enhances the interpretability of recommendations by disentangling embedding vectors of user interests from herd behavior.
 - e) *PDA* [37] eliminates the adverse effects of popularity bias during the model training phase. It then introduces the desired popularity through causal intervention in the inference phase, thereby improving the accuracy of recommendations.
 - f) *BERT4Rec* [53] is a recommendation model that leverages deep bidirectional self-attention mechanisms. It is trained on Cloze tasks and predicts masked items within user behavior sequences to achieve personalized recommendations.
 - g) *DT4SR* [54] is a Transformer-based sequential recommendation model. It captures the relationship between users' dynamic preferences and items by modeling user interaction sequences as Gaussian distributions with uncertainties.

- h) *STOSA* [42] is also a Transformer-based sequential recommendation model. It represents items as random Gaussian distributions and introduces an innovative Wasserstein self-attention mechanism. Additionally, it proposes a novel regularization term to optimize the ranking loss.

3) *Implementation Details*: For the above datasets, we consider the presence of ratings as positive interactions. To facilitate more accurate and reliable model training and evaluation, we normalize the rating data to a consistent range of [0, 1]. Based on our previous analysis of the auxiliary variable B_t , we calculate the item temporary popularity and its Pearson correlation coefficient with rating changes. Together with the item ratings, they are used to disentangle the beneficial and harmful effects of popularity. To capture users' personalized demand factors, we obtain and normalize the basic matching scores between users and items using the classic backbone model BPR-MF. For the user-item interaction sequences U_t and I_t , we sort the interaction data by timestamps and construct them using multihot matrices, where interactions are marked as 1 and noninteractions as 0. Following the model structure described in the methodology section, we concatenate these variables and input them into the prior and posterior networks to obtain the distributions of hidden variables, which are then used to generate the final prediction results.

Additionally, we divide the dataset into five equal time intervals in chronological order. Within each interval, we employ the leave-one-out evaluation task widely used in [20], [42], [53]. For each user's sequential interaction sequence, the most recent interaction serves as test data, the second-to-last interaction as validation data, and the remaining items are used for training. To ensure the reliability of user evaluation, we apply a 10-core filtering to the KuaiRec and Douban-Movie datasets, ensuring that each user has at least ten interaction records with items. Given the high sparsity of the Amazon-CDs dataset, we implement a 5-core filtering for it.

During the training process, the parameters of the observational variable extraction model are randomly initialized and iteratively optimized. We employ the Adam optimizer to estimate hyperparameters, setting the weight decay to 0.0001, the learning rate to 0.001, and the embedding size to 64. To ensure fairness in our experimental results, we set different embedding dimensions based on the varying structures of the baseline models. Considering that PDA, DICE, and STOSA models contain two sets of embeddings, we search for their embedding dimensions within {32, 64}. Other baseline models search within {64, 128}. Additionally, following the relevant work in [37], [38], [42], [53], we optimally set parameters such as the maximum sequence length, learning rate, L2 regularization weight, and drop rate for the baseline models, employing grid search for hyperparameter tuning.

To prevent overfitting across all models, we limit the number of epochs. Training is halted when there are ten consecutive performance declines on the validation set. The experiments are implemented in PyTorch, with an Intel i7 CPU and an Nvidia 3090 GPU.

4) *Evaluation Metrics*: We conduct experiments to evaluate the performance of the CDSRec model we proposed. Traditional recommendation systems prediction tasks primarily use static metrics such as Recall@K, Precision@K, and normalized discounted cumulative gain (NDCG@K). These metrics focus solely on the performance of the recommendation system at a specific point in time [55]. However, in scenarios where user preferences and item popularity change dynamically over time, it's crucial to assess the model's performance across different time periods. To address this, we propose a dynamic evaluation method to better capture the performance of recommendation systems along the temporal dimension. The specific metrics are described as follows:

Time-Aware NDCG@K ($NDCG_T@K$) measures the ranking quality of recommendation results, taking into account the relevance and position of recommended items. We introduce a temporal weighting mechanism, allowing predictions at different times to have differentiated impacts on the recommendation results. The formula is as follows:

$$DCG_t@K = \sum_{i=1}^K \frac{2^{rel_i} - 1}{\log_2(i + 1)} \quad (12)$$

$$NDCG_T@K = \frac{1}{|T|} \sum_{t=1}^T \frac{DCG_t@K \cdot \omega(t)}{IDCG_t@K} \quad (13)$$

where T represents the number of time intervals, $DCG_t@K$ denotes the Discounted Cumulative Gain within time interval t , and $NDCG_T@K$ is the maximum DCG value for the ideal recommended sequence. $\omega(t)$ is the weight function at time t , which increases over time. $rel_i \in \{0, 1\}$ indicates the relevance of the i th recommended item, where the value is 1 if there is an interaction, and 0 otherwise.

Dynamic Recall@K ($Recall_T@K$) takes into account the temporal dimension, aiming to assess the variation in prediction accuracy of the model across multiple time intervals. It can be used to measure the proportion of successfully recommending relevant items within the top K recommended items, defined as

$$Recall_T@K = \frac{1}{|T|} \sum_{t=1}^T \frac{|L_t \cap P_t@K|}{|L_t|} \cdot \omega(t) \quad (14)$$

where L_t denotes the set of items that users actually interacted during the time interval t , and $P_t@K$ represents the top K predicted items by the model. By calculating the weighted average of the recall rate across multiple time intervals, $Recall_T@K$ can evaluate the overall performance of the recommendation model over different times.

These metrics encompass key dimensions such as prediction accuracy and coverage, providing a comprehensive evaluation of the recommendation model. Higher values indicate better model performance.

B. Analysis of Experimental Results

To evaluate the recommendation performance of our proposed CDSRec model, we conduct experiments across three datasets and compared the results with the baseline models mentioned above. The overall recommendation performance

TABLE II
OVERALL PERFORMANCE OF THE PROPOSED METHOD AND BASELINE MODELS ON THREE DATASETS

	KuaiRec				Douban-Movie				Amazon-CDs			
	Top 20		Top 50		Top 5		Top 20		Top 5		Top 20	
	Recall	NDCG	Recall	NDCG	Recall	NDCG	Recall	NDCG	Recall	NDCG	Recall	NDCG
BPR-MF	0.0059	0.0061	0.0122	0.0088	0.0098	0.0260	0.0272	0.0421	0.0059	0.0065	0.0117	0.0090
LightGCN	0.0088	0.0075	0.0159	0.0112	0.0131	0.0305	0.0304	0.0450	0.0086	0.0085	0.0165	0.0117
CausE	0.0079	0.0084	0.0129	0.0137	0.0133	0.0285	0.0278	0.0409	0.0075	0.0071	0.0135	0.0112
DICE	0.0062	0.0079	0.0137	0.0122	0.0150	0.0314	0.0283	0.0457	0.0067	0.0079	0.0138	0.0118
PDA	<u>0.0146</u>	0.0138	<u>0.0271</u>	0.0206	0.0204	0.0406	0.0406	0.0596	0.0105	0.0110	<u>0.0227</u>	<u>0.0168</u>
BERT4Rec	0.0101	0.0114	0.0179	0.0159	0.0149	0.0325	0.0310	0.0436	0.0095	0.0092	0.0182	0.0124
DT4SR	0.0120	<u>0.0143</u>	0.0213	0.0196	0.0186	0.0384	0.0355	0.0501	0.0112	0.0103	0.0195	0.0148
STOSA	0.0135	0.0141	0.0255	<u>0.0213</u>	<u>0.0220</u>	<u>0.0460</u>	<u>0.0423</u>	<u>0.0625</u>	<u>0.0122</u>	<u>0.0118</u>	0.0223	0.0156
Ours	0.0162	0.0167	0.0306	0.0234	0.0237	0.0491	0.0445	0.0639	0.0129	0.0127	0.0236	0.0172
Improv.	10.96%	16.78%	12.92%	9.86%	7.73%	6.74%	5.20%	2.24%	5.74%	7.63%	3.96%	2.38%

Note: Bold indicates the best experimental outcomes of different models under the same experimental setup and evaluation metrics, with underlined data representing the second-best results, and “Improv.” referring to the improvement percentage based on these two.

is presented in Table II. Based on the results, we obtain the following analysis:

1) *Performance Comparison (RQ1)*: Compared to the baseline models, CDSRec achieves the best results across all metrics. It particularly excels on the KuaiRec dataset, with relative improvements ranging from 9.86% to 16.78% in all metrics. Additionally, the model demonstrates strong adaptability and robustness in handling items across diverse datasets, effectively addressing the challenges posed by varying degrees of popularity bias. We attribute these improvements to the following factors: a) introducing the causal disentanglement mechanism: Based on causal inference and hidden Markov model, CDSRec separates key hidden variables that influence user preferences (such as personalized demand and inherent quality) from external interference factors, effectively reducing the adverse effects of popularity bias; b) modeling temporal dependencies: CDSRec utilizes time-series analysis to extract valuable information from attribute data and historical interaction sequences, effectively and promptly capturing the dynamic changes in item popularity and user preferences; and c) optimizing network frameworks: By reformulating the objective function, we ensure that the model can be trained and predicted according to our specific setup.

Traditional static methods, including BPR-MF and LightGCN, often serve as baseline models for comparison in recommendation systems. These models primarily focus on the static relationships between users and items, neglecting the influence of time factors, which leads to their inferior results compared to sequential recommendation approaches. This outcome confirms the necessity of incorporating temporal information in recommendations. Compared to BPRMF, LightGCN performs better, likely due to its graph convolutional network model, which more effectively captures the local connection patterns between users and items. This enhances the embedding representations of users and items, leading to more accurate recommendations.

Existing causal inference-based methods for addressing popularity bias, including CausE, DICE, and PDA, generally outperform static methods. This is especially evident in datasets with severe popularity bias, such as short video interaction platforms. In particular, PDA stands out among the baseline models by employing causal intervention techniques to eliminate confounding bias, and introducing desired popularity bias during the recommendation process. This result indirectly confirms the presence of item popularity bias and suggests that such bias is not solely detrimental.

Among all the sequential recommendation baselines, STOSA achieved the best performance. Compared to traditional transformer-based self-attention methods such as SASRec and BERT4Rec, STOSA represents each item as a random Gaussian distribution. By introducing a novel Wasserstein self-attention mechanism, it enhances recommendation performance. However, in datasets with dense interaction data and significant popularity bias, existing sequential recommendation methods may decrease user satisfaction due to their failure to account for the influence of popularity factors.

2) *Effectiveness of Disentanglement (RQ2)*: To demonstrate the effectiveness of disentanglement in our model, we design ablation studies and conduct further validation of the hidden variables x , y , and z on the Douban dataset. Initially, we train the CDSRec model normally to obtain the disentangled distributions of the hidden variables. Subsequently, we train six independent classifiers, each using different combinations of the hidden variables for prediction. The experimental results are presented in Table III.

We investigate whether eliminating external interference factors during the prediction stage is beneficial. Building on the original prediction model, we incorporate the hidden variable z which represents external interference factors. Compared to the standard model, the prediction results including z show a decrease of 8.4% and 6.2% in Recall metrics, and a decrease

TABLE III
PREDICTION RESULTS OF DIFFERENT HIDDEN VARIABLES

Variables			Douban-Movie			
x	y	z	Recall@5	NDCG@5	Recall@20	NDCG@20
×	✓	×	0.0120	0.0208	0.0223	0.0402
×	✓	✓	0.0098	0.0190	0.0208	0.0376
✓	×	×	0.0225	0.0462	0.0426	0.0614
✓	×	✓	0.0185	0.0363	0.0360	0.0537
✓	✓	✓	0.0213	0.0454	0.0417	0.0596
✓	✓	×	0.0237	0.0491	0.0445	0.0639

Note: Bold indicates the optimal results of ablation experiments under the same evaluation metrics.

of 7.5% and 6.7% in NDCG metrics, respectively. These results confirm the presence of external interference factors in recommended items, suggesting that incorporating these factors can negatively impact recommendation performance.

We attempt to completely eliminate popularity bias by focusing solely on the personalized demand factors (i.e., hidden variable x) for recommendations, without considering the inherent quality of items. The results show that recommendations using only x perform worse than those utilizing both x and y . Notably, when K is set to 5, there is a 5.9% reduction in NDCG and an 5.1% reduction in Recall. The experiment demonstrates that popularity bias is not entirely detrimental, as popular items often have higher quality. Appropriately incorporating popularity bias can improve recommendation performance.

We also conduct additional combination experiments. For instance, in scenarios where user historical interaction data is lacking, we remove the personalized demand factors x and use other hidden variables for recommendations. The results indicate that focusing solely on the positive effects of popularity (introducing y) performs slightly better than incorporating external interference factors (introducing y and z). However, both of these attempts perform significantly worse than our proposed standard model. This highlights the importance of considering user personalized demand factors and further confirms the effectiveness of disentanglement in our model.

3) *Impact of Temporal Information (RQ3)*: To explore the significance of temporal information in addressing the problem of popularity bias, we design an experiment. Initially, we analyze the variation cycles of item popularity. Considering the shorter time span of the KuaiRec dataset, we select it for our experiment to facilitate evaluation. A random subset of items is selected and divided into equal time intervals in days according to their chronological order. Based on (1), we calculate the temporary popularity metrics for items within each interval. For different coefficient ranges (including positive, negative, and zero values), we select three representative items to illustrate the variation in their temporary popularity over time, as shown in Fig. 5. We observe that items with positive coefficients tend to have a relatively smooth and sustained popularity. In contrast, items with negative coefficients often exhibit a rapid spike in popularity, followed by a gradual decline and eventual stabilization. Items with coefficients close to zero fall somewhere

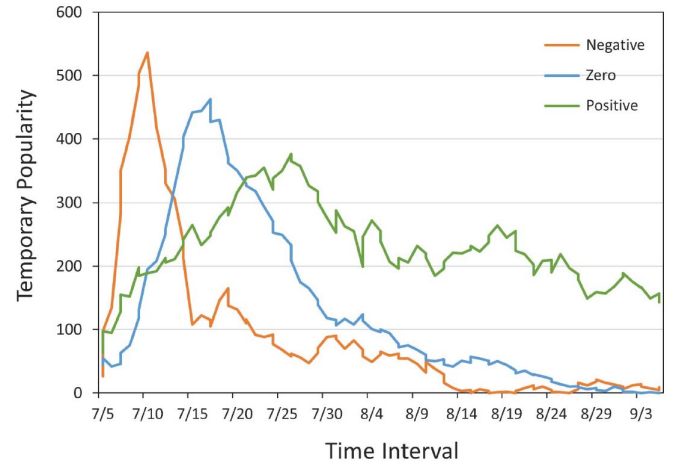


Fig. 5. This graph illustrates the change in item temporary popularity across different time intervals, with three items selected to represent varying coefficient ranges.

between them. We statistically analyze the temporary popularity sequences of each item and set the 70% of the maximum value as the threshold. Periods exceeding this threshold are considered to be within the popular interval. We calculate that the average duration of the popularity intervals across all items is approximately 8 days.

Based on the above analysis, to ensure that most items experience changes in popularity during the training period, we divide the dataset into six equal time intervals according to the interaction timestamps, denoted as N . Each time interval spans 10 days. We set up six different experimental sequence lengths; for example, “N1 to N6” means the experiment starts from the first time interval and ends at the sixth. Within each interval, we employ the leave-one-out evaluation method, with experimental settings referring to the previous section. We select the best-performing methods from different model categories for comparison, analyzing the recommendation performance across different time intervals.

The experimental results are shown in Table IV. For a more intuitive presentation, we also plot the results in a line chart as shown in Fig. 6. We observe that as the time intervals extend, sequential recommendation models generally outperform other models. The improvement is particularly significant when the time intervals are shorter, likely due to the sequential recommendation model’s better ability to leverage time series information. The increase in the time intervals can substantially cover the variation cycles of item popularity and ratings, allowing us to discern the presence of popularity bias and thereby mitigate its adverse effects. The results of the CDSRec model surpass the current state-of-the-art STOSA method, thereby demonstrating the effectiveness of our model in addressing the popularity bias in sequential recommendation. When the time intervals are sufficiently long, the improvement becomes less pronounced, possibly because longer windows already contained sufficient interaction information. Further increasing the sequence length may introduce computational complexity and additional interference factors, negatively impacting the model’s predictions.

TABLE IV
RECOMMENDATION PERFORMANCE ACROSS DIFFERENT TIME INTERVALS ON THE KUAIREC DATASET

	LightGCN		PDA		STOSA		CDSRec	
	Recall@50	NDCG@50	Recall@50	NDCG@50	Recall@50	NDCG@50	Recall@50	NDCG@50
N6 to N6	0.0165	0.0115	0.0270	0.0197	0.0256	0.0205	0.0286	0.0219
N5 to N6	0.0169	0.0120	0.0275	0.0203	0.0268	0.0212	0.0303	0.0228
N4 to N6	0.0171	0.0128	0.0279	0.0209	0.0276	0.0219	0.0316	0.0239
N3 to N6	0.0173	0.0122	0.0276	0.0212	0.0280	0.0225	0.0324	0.0246
N2 to N6	0.0170	0.0127	0.0278	0.0206	0.0286	0.0229	0.0327	0.0250
N1 to N6	0.0169	0.0125	0.0278	0.0203	0.0290	0.0227	0.0329	0.0251

Note: Bold indicates the best results for each, reflecting the impact of changes in temporal information on the outcomes.

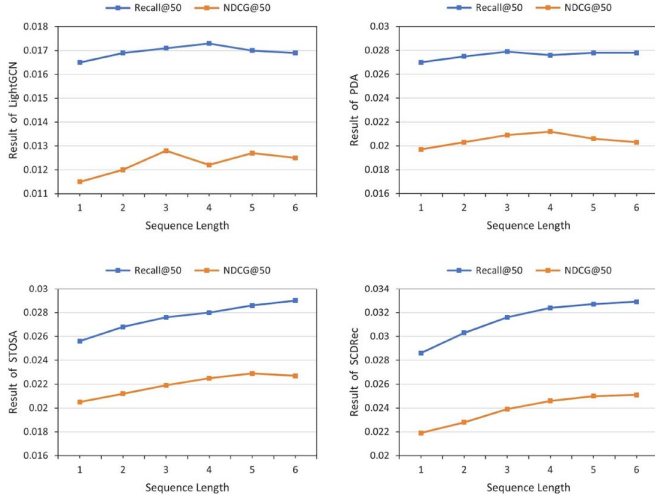


Fig. 6. Line chart of comparative recommendation results across different model categories.

Additionally, we find that the static recommendation model LightGCN and the causal inference model PDA perform modestly on long-sequence recommendation tasks. As the time window increases, the improvements in both Recall and NDCG are relatively small. These models focus more on interaction information close to the prediction time. Particularly for LightGCN, when the time interval become longer, the predicted results tend to fluctuate and decline.

VI. CONCLUSION

In this article, we propose the CDSRec, a novel approach to tackle the challenges of popularity bias in dynamic scenarios. By conducting time-series analysis on the historical user-item interaction sequences and attribute information, we capture the dynamic changes in item popularity and user preferences. Based on causal inference and hidden Markov model, we disentangle the factors that contribute to popularity bias. We train our model by optimizing the sequential VAE framework, which reduces the adverse effects of external interference factors and extracts beneficial information for recommendations. Extensive experiments on three typical datasets demonstrate the improvement in recommendation performance compared to existing methods. For future research, we plan to extend the model to a

wider range of recommendation scenarios. Processing data at a more granular level to enhance the ability to make sequential decisions.

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